INSTRUCTIONS TO THE CANDIDATES

- Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
- Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
- Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
- Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
- The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.

  Non-adherence with this instruction shall result in candidate being booked under malpractice.
- Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
- Candidate should write only question number in the margin.
- Candidate should return the answer booklet to the invigilator before leaving the examination hall.
- **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

INSTRUCTIONS TO THE EXAMINERS

- Invigilators should ensure that the above instructions are strictly followed by the candidates.
- After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- Do not crumple or cut the sides of the sheets.
- Do not write, mark or damage the timing tracks or Barcode.

5

5.B 8051 Program.

Copy a block of 10 bytes from 40H to 50H.

```
ORG 00H
MOV R0, 40H
MOV R1, 50H
MOV R2, 11H
Back: MOV A, @R0
      MOV @R1, A
      INC R0
      INC R1
      DJNZ R2, Back.
```

5.A Logical instructions.

ANR
OR
XOR
NOT
NAND

AND :- it performs the and operation on 2 ~~objects~~ register which of passed to it.

AND R_1 , R_2

OR :- It performs the or operation between 2 ~~objects~~ register which of passed to it

OR R_1 , R_2.

XOR. :- It performs the exclusive or operation betwe 2 registes passed to it

XOR R_1 , R_2

NOT :- It performs the complementary operation on the register passed to it

NOT R_1

NAND :- It performs the complementary and operation ~~on~~ between 2 registers passed to it

2

2.A Signals

i) INTR

Interupt register. → Input to the board

ii) ALE - Arithmetic Logic Enable
Output.

iii) HOLD
Input.

2.B. 8086 addressing modes

Direct addressing mode

When the source is a register and the destination is a register

~~MOV AR1, R2~~
MOV A_x, B_x

Register indirect addressing mode.

When the source is given indirectly with ~~the~~ an address which points to a register which points to the data.

MOV A_x, [Rx]

2.C 8051 Addressing modes

Immediate Addressing mode

When the source is directly provided with the immediate data value.

MOV R_1, 10H

Direct Addressing mode
When the source is provided with a direct register of the value

MOV R_1, R_3

2.D

There are a total of 4 register banks.
we have 2 ~~gi~~ registers which decide which register bank is to be selected

RB0	RB1	Bank
0	0	bank 0
0	1	bank 1
1	0	bank 2
1	1	bank 3

To give the registers the values of 0 or 1, we use SETB and CLEAR.

starting - 10H
ending - 1FH

1

1.A

20 bit physical address bus in 8086

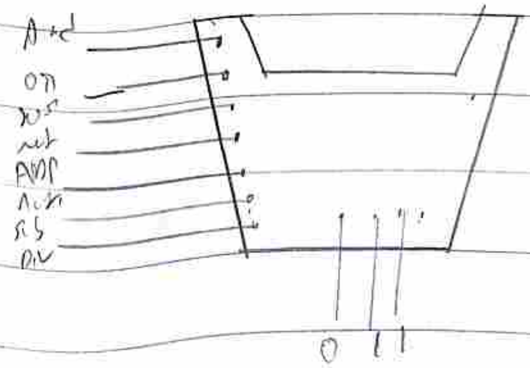
20 bit is 2.5 bytes.

The first 8 bits is 1 byte
The next 8 bits is another 1 byte
The next 4 bits is a half byte.

Some times the address bus is recognised as a 3 bytes

1.B ALU

Arithmetic logic unit



Arithmic logic units, will give a function either Arithmaical or logical depending on the inputs given to it

1.D ~~Segment/register~~.

AX → Arithmatic register
CX → code register

1.E IE (Interrupt Enable)

In every micro controller there will be intrrupts
The order of proirity of intrrupts is IE

4
4.A Instructions

1.) OR AX, AX

You shouldn't use the same register twice in OR

ii) AND AX, [OABCDH]

You should used ~~registers~~ proper Address to denote in function.

iii) RCL AL, 01H
iv) RCR Al, 01H

V.) RCL
VI.) RCR

4.B

```
ORG 100H
mov Ax, [0430H]
Mov Bx, 0431H
mov C, [Ax]
Mov D, [Bx]
mul CD
clear Ax
mov Ax, 1200h
mov [Ax], C
```

0		1		2	
16		16		16	
00	08	10	18	20	28
01	09	11	19	21	29
02	0A	12	1A	22	2A
03	0B	13	1B	23	2B
04	0C	14	1C	24	2C
05	0D	15	1D	25	2D
06	0E	16	1E	26	2E
07	0F	17	1F	27	2F